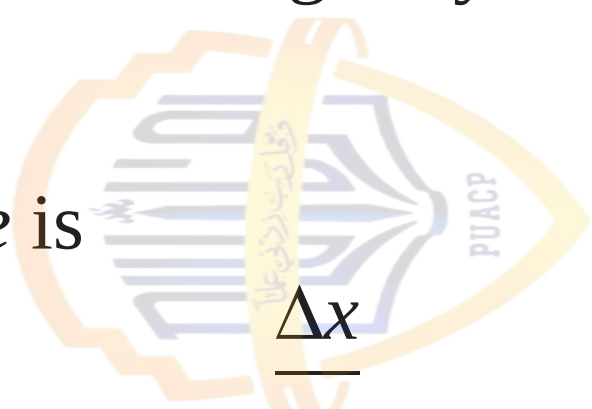


Fractional Change

Consider a variable x that changes by Δx during a time period of length Δt .

The *fractional change* is


$$\frac{\Delta x}{x}$$

For example, a fractional change of .05 would mean an increase of 5%.

Growth Rate

The *growth rate* is the fractional change per unit time,

$$\frac{\frac{\Delta x}{x}}{\Delta t},$$

the fractional change divided by the length of the time period.

For example, suppose that the length of the time period is one quarter year, $\Delta t = .25$. And suppose that the fractional change in the variable is .02. Then the growth rate is

$$\frac{.02}{.25} = .08,$$

a rate of 8% per year.

Growth Rate in Continuous Time

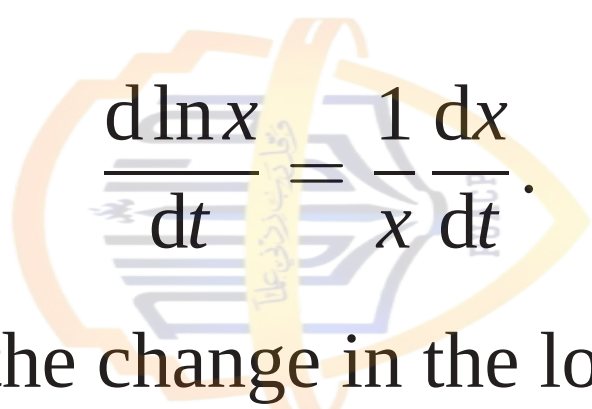
The growth rate in continuous time is the growth rate as the length of the period shrinks to zero, $\Delta t \rightarrow 0$.

In the limit, the growth rate is the derivative,

$$\lim_{\Delta t \rightarrow 0} \frac{1}{x} \frac{\Delta x}{\Delta t} = \frac{1}{x} \frac{dx}{dt}.$$

Derivative of the Logarithm

The growth rate is also the derivative of the logarithm,


$$\frac{d \ln x}{dt} = \frac{1}{x} \frac{dx}{dt}.$$

For a small changes, the change in the logarithm must be the fractional change,

$$\Delta \ln x \approx \frac{\Delta x}{x}.$$

Constant Growth Rate

If the variable grows at the constant rate g ,

$$\ln x_t = \ln x_0 + gt.$$

Taking the exponent shows

$$x_t = e^{\ln x_t} = e^{(\ln x_0 + gt)} = e^{\ln x_0} e^{gt} = x_0 e^{gt}.$$

Thus

$$e^{gt}$$

represents constant growth at rate g .

Graph

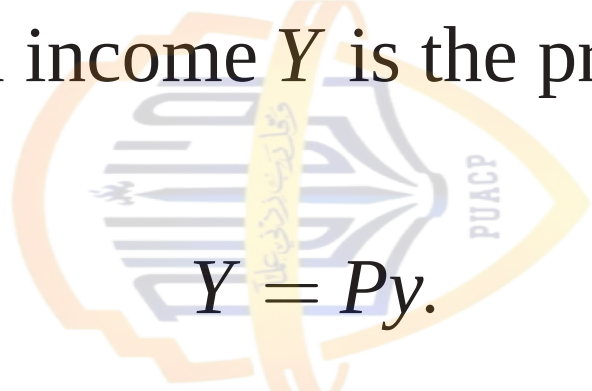
A non-logarithmic graph of $x_t = x_0 e^{gt}$ against time has a steadily increasing slope.

A logarithmic graph of $\ln x_t = \ln(x_0 e^{gt}) = \ln x_0 + gt$ against time has the constant slope g . Equal upward and downward shifts along the curve represent equal upward and downward percentage changes in the variable.

Growth Rate of a Product

The growth rate of a product is the sum of the growth rates.

For example, nominal income Y is the price level P times real income y ,


$$Y = Py.$$

Then the growth rate of nominal income is the inflation rate plus the growth rate of real income,

$$\begin{aligned}\frac{1}{Y} \frac{dY}{dt} &= \frac{d \ln Y}{dt} \\ &= \frac{d \ln Py}{dt} \\ &= \frac{d (\ln P + \ln y)}{dt} \\ &= \frac{d \ln P}{dt} + \frac{d \ln y}{dt} \\ &= \frac{1}{P} \frac{dP}{dt} + \frac{1}{y} \frac{dy}{dt}.\end{aligned}$$

Growth Rate of a Ratio

Analogously, the growth rate of a ratio is the growth rate of the numerator minus the growth rate of the denominator.

For example, the growth rate of the real wage $\frac{W}{P}$ is the growth rate of the nominal wage W minus the growth rate of the nominal price level P .